

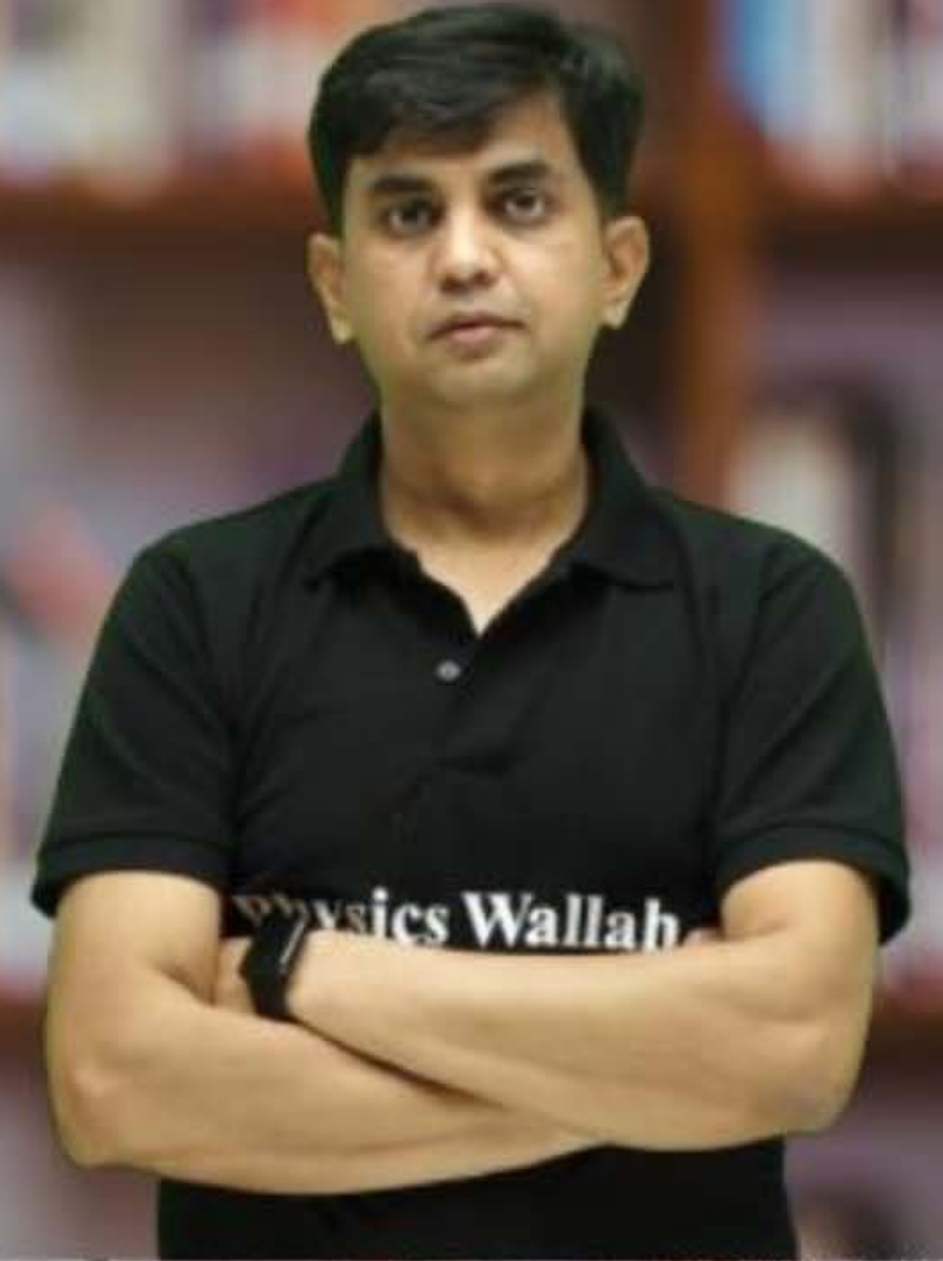
CS & IT ENGINEERING



Computer Network

Socket API

Lecture No. - 03



By - Abhishek Sir



Recap of Previous Lecture



Topic

TCP Connection Close





Topics to be Covered



Topic

Socket API



ABOUT ME

Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

My Telegram Channel

Telegram Link : https://t.me/abhisheksirCS_PW





Topic : Network Socket

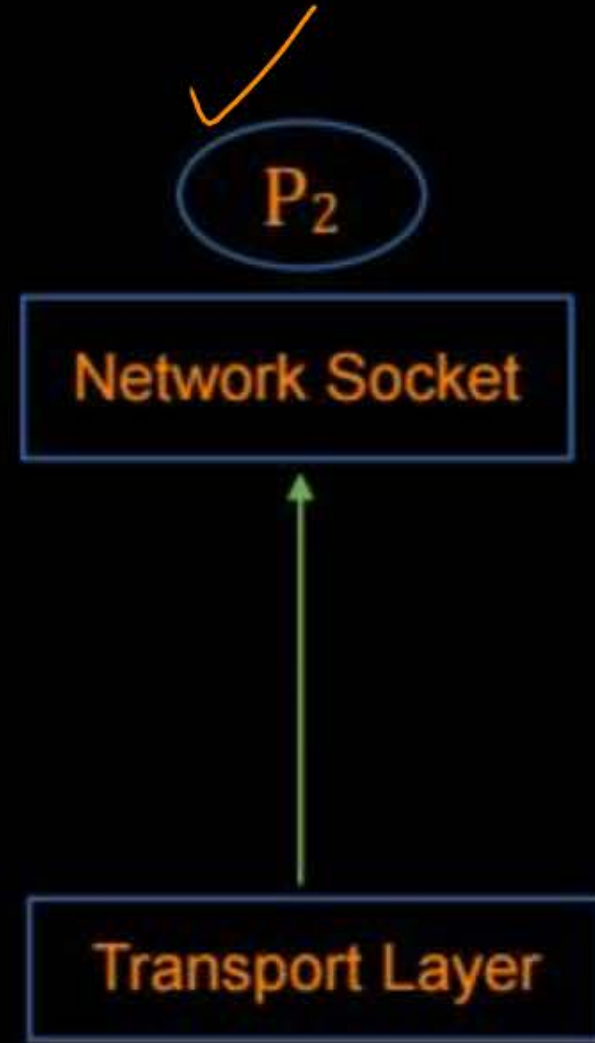


→ Door or interface of a network process

→ Data structure / object



Topic : Network Socket





Topic : Network Socket



→ Two network process communicate via a pair of sockets

→ One endpoint of a two-way communication



Topic : Network Socket



Two Transport Layer Protocols :

1. UDP Socket

Datagram Socket : SOCK_DGRAM

2. TCP Socket

Stream Socket : SOCK_STREAM

↳ byte stream



Topic : UDP Socket





Topic : TCP Socket





Topic : TCP



TCP Client Process

P₁

Port No. A

SOCK_STREAM
(TCP Socket)

TCP Connection Establishment

3-way handshake

TCP Server Process

P₂

Port No. B

SOCK_STREAM
(TCP Welcome
Socket)

Fork()

child
process

P₂₁



Topic : TCP



TCP Client Process

P_1

Port No. A

SOCK_STREAM
(TCP Socket)

TCP Server Process

P_2

Port No. B

SOCK_STREAM
(TCP Welcome
Socket)

P_{21}

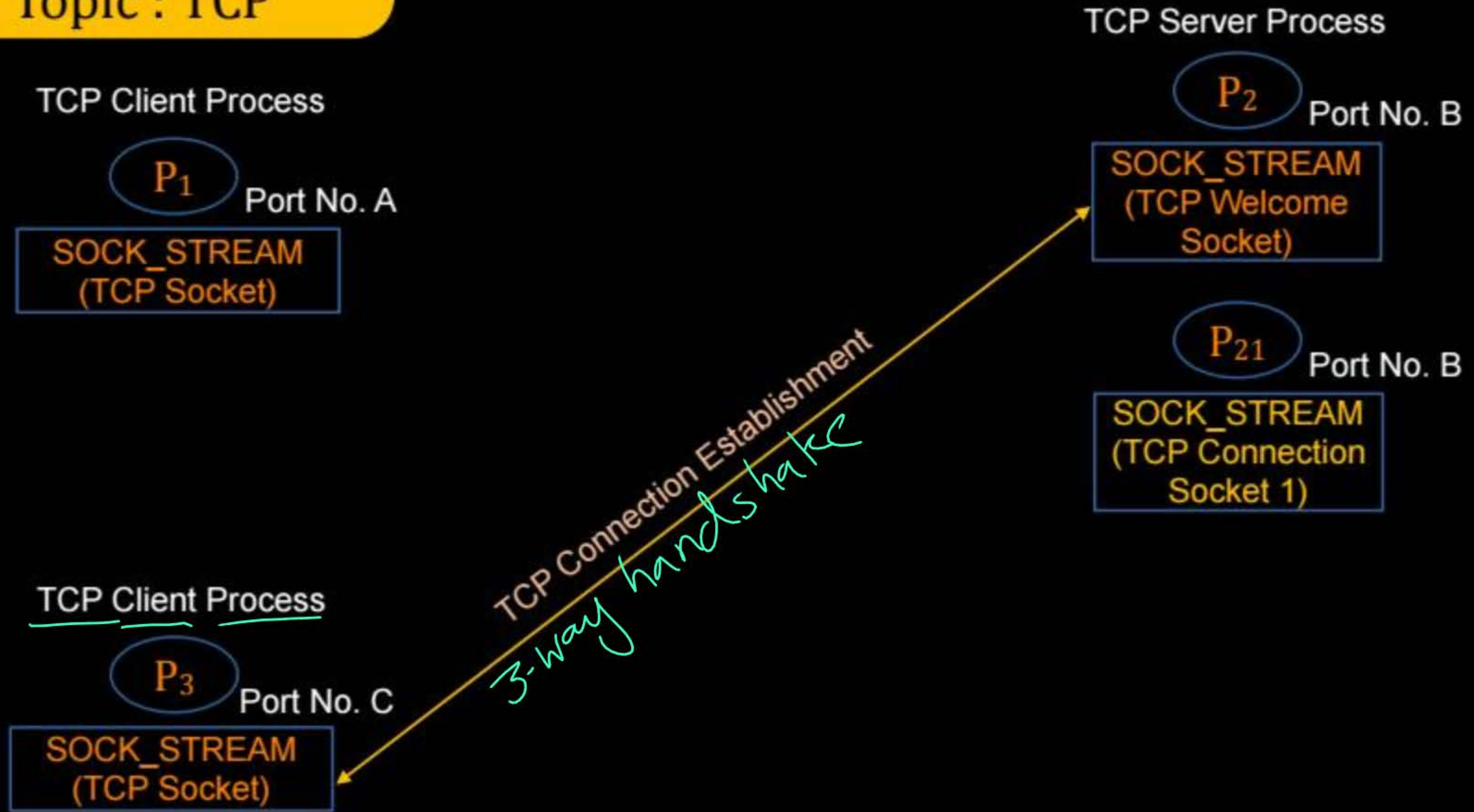
Port No. B

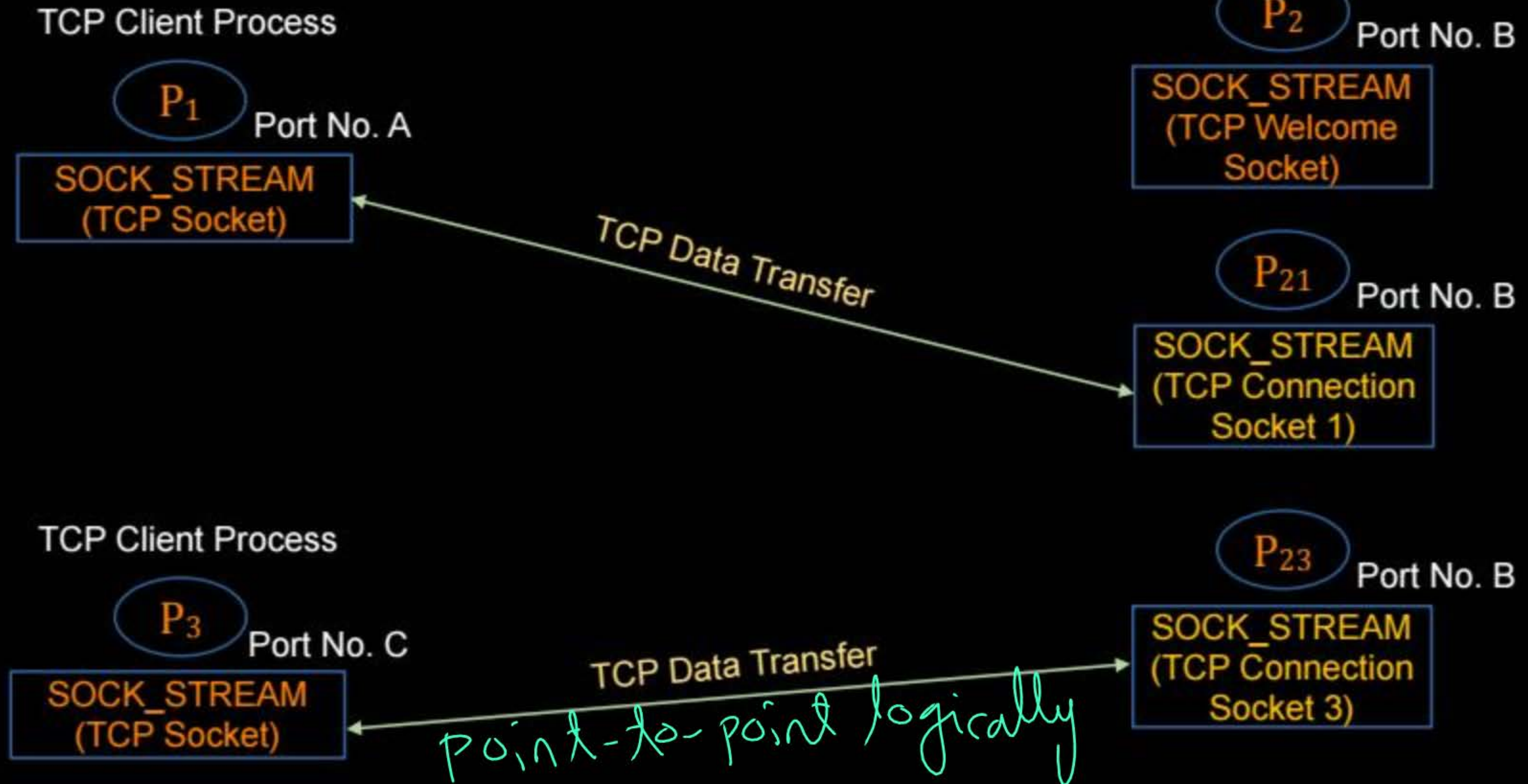
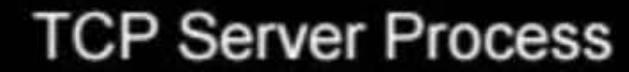
SOCK_STREAM
(TCP Connection
Socket 1)

(TCP Data Transfer)
*Point-to-point
logical connected;*



Topic : TCP







Topic : Connection-Oriented Demultiplexing

→ TCP sockets identified by 4 tuples :

Destination IP address and Destination Port Number

Source IP address and Source Port Number



Topic : TCP



TCP Client Process

P_1

SOCK_STREAM
(TCP Socket)

Send Buffer

Recv Buffer

*logically
point-to-point*

TCP Server Process

P_{21}

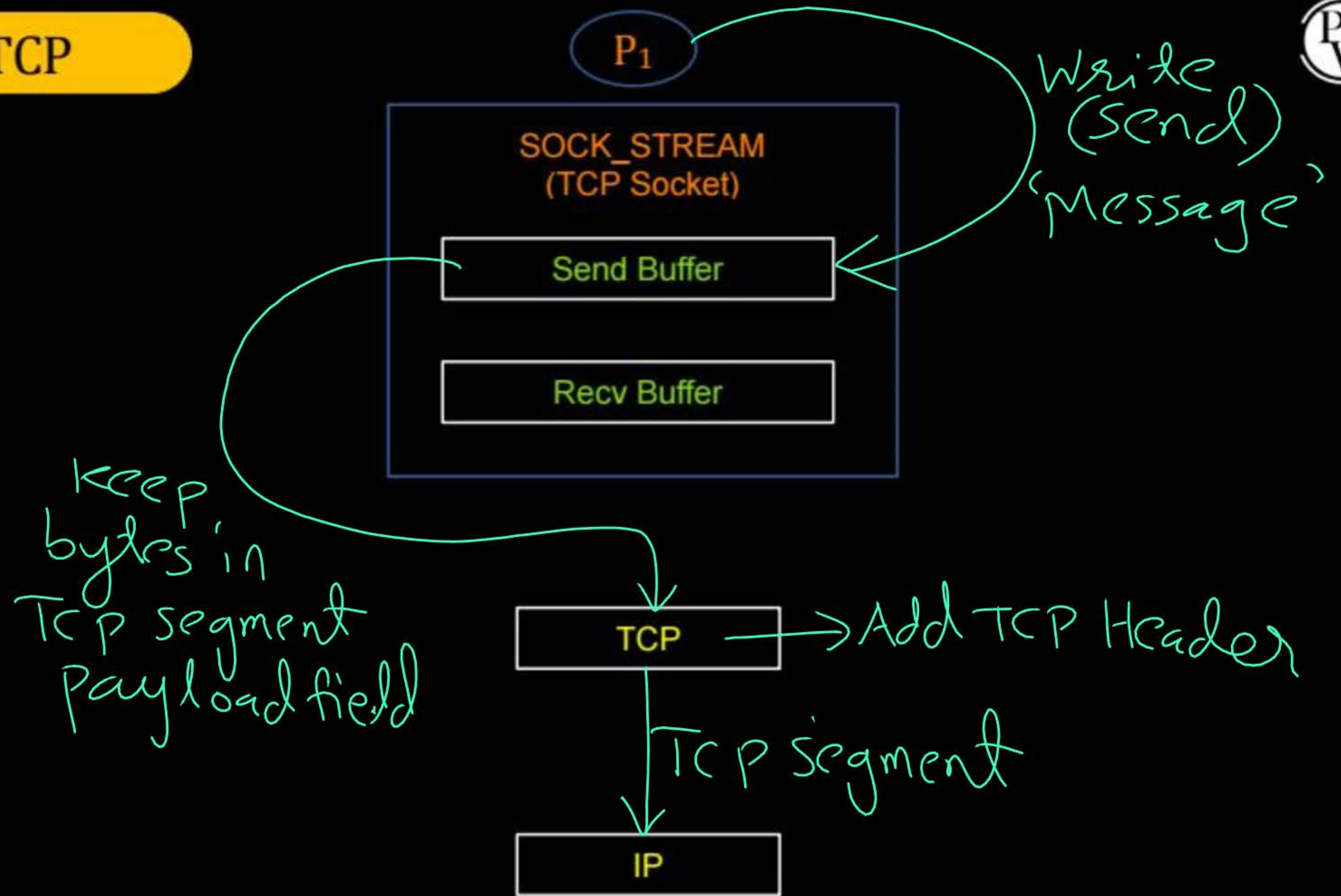
SOCK_STREAM
(TCP Connection Socket 1)

Send Buffer

Recv Buffer



Topic : TCP

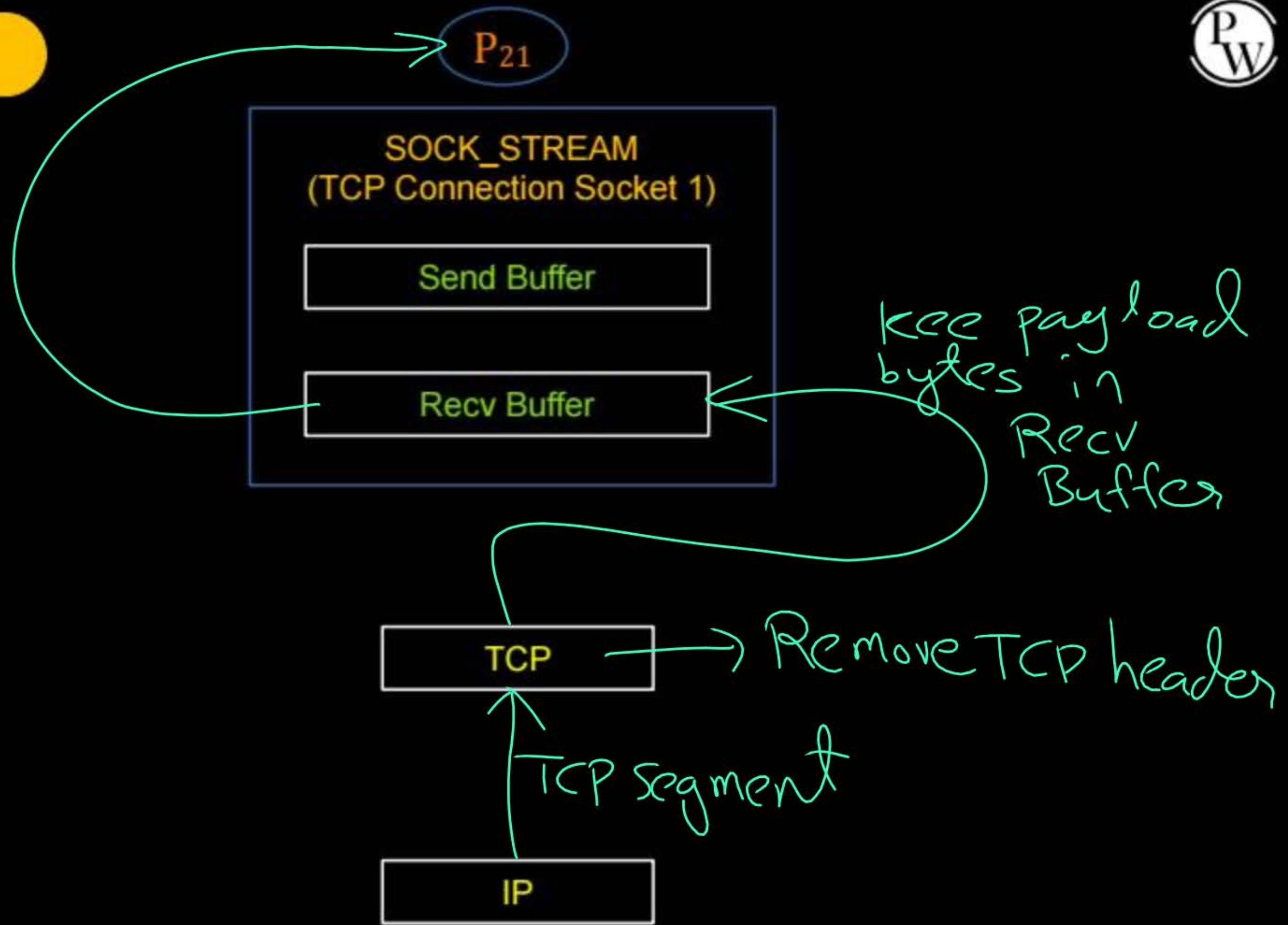




Topic : TCP



Read
(Recv)
'Message'





Topic : Socket API



→ Set of library function calls

[Network (Socket) Communication]

→ socket programming
→ Network programming

→ API : Application Programming Interface



Topic : Types of Sockets

Stream Socket : SOCK_STREAM

- Uses by TCP protocol
- Byte-stream oriented
- For connection-oriented and reliable service



Topic : Types of Sockets



Datagram Socket : **SOCK_DGRAM**

- Uses by UDP protocol
- Connection-less and unreliable service



Topic : Creating a Socket



```
int socket ( int domain, int type, int protocol );
```

→ TCP/UDP

→ On success, file descriptor of newly created socket;

→ on error, -1 return;



Topic : Creating a Socket



type : Communication Protocol semantics

[SOCK_STREAM, SOCK_DGRAM]

↓
TCP

↓
UDP



Topic : Bind a name to a Socket

```
int bind ( int sockfd, const struct sockaddr *addr, socklen_t addrlen );
```

→ on success, 0 return;

→ on error, -1 return;



Topic : Listen for connections

Max^m length of Accept queue;

```
int listen ( int sockfd, int backlog );
```

Listening/welcome socket

* Listen state, TCP server

* Ready to read client req.

0 }
-1 } ignore



Topic : TCP Socket Programming



P₁
TCP client
create socket ← socket()

bind is optional at client

P₂
TCP Server
socket() → create socket
bind() → Address binding
listen() → welcome socket



Topic : Initiate a connection



```
int connect ( int sockfd, const struct sockaddr *addr, socklen_t addrlen );
```

- * After some time before timeout SYN-ACK Recv. (on success, return 0;)
- * No SYN-ACK recv. till timeout (on error, -1 return)



Topic : TCP Socket Programming

P_1
TCP Client

socket()

connect()

P_2
TCP Server

socket()

bind()

listen()

*←

Accept Queue

SYN

TCP Connection Req.

[MCQ] [GATE-CS-2008] [1-Mark] 115C



#Q. Which of the following system calls results in the sending of SYN packets?

- ☐ A Socket
- ☐ B Bind
- ☐ C Listen
- ☒ D Connect

Ans: D



Topic : TCP Socket Programming

P_1
TCP Client

socket()

connect()

P_2
TCP Server

socket()

bind()

listen()

*

Accept queue



return 0;
(Success)
Connection
Established;



Topic : Accept a connection

welcome/listen socket

int accept (int sockfd, struct sockaddr *_Nullable restrict addr,
socklen_t *_Nullable restrict addrlen);

→ on success, new socket file descriptor return;
* fork(), child process

→ on error, -1 return;



Topic : TCP Socket Programming

(3-way Handshake)



P₁
TCP Client

socket()

connect()

P₂
TCP Server

socket()

bind()

listen()

accept()

Accept Que

new socket
P₂₁

⇒ welcome socket

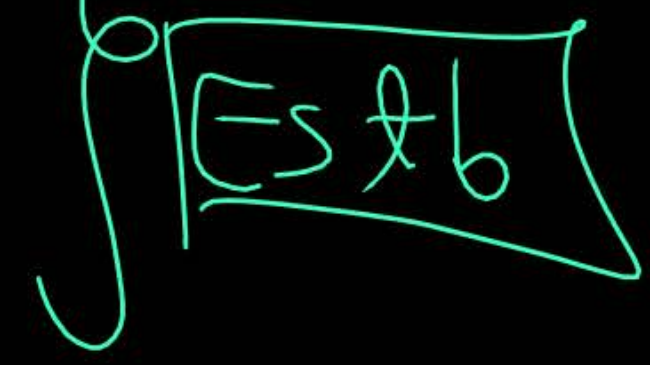
SYN

SYN - ACK

(ACK)

Estb.

Estb





Topic : TCP Socket Programming

P_1
TCP Client

socket()

connect()

P_2
TCP Server

socket()

bind()

listen() → Request block;
(Accept Queue
is full)

SYN

Time-out
Return -1;
Error



Topic : Write to a Socket



```
ssize_t write ( int sockfd, const void buf[count], size_t count);
```

To write (send) in send buffer by process



Topic : Read from a Socket



```
ssize_t read ( int sockfd, const void buf [.count], size_t count );
```

* Read (Recv) from receive buffer by process.

[MCQ] [GATE-2015, Set-2] [1-Mark]

IIT-K, H.W. (Telegram)



#Q. Identify the correct order in which a server process must invoke the function calls accept, bind, listen, and recv according to UNIX socket API.

- ☐ A listen, accept, bind, recv
- ☐ B bind, listen, accept, recv
- ☐ C bind, accept, listen, recv
- ☐ D accept, listen, bind, recv



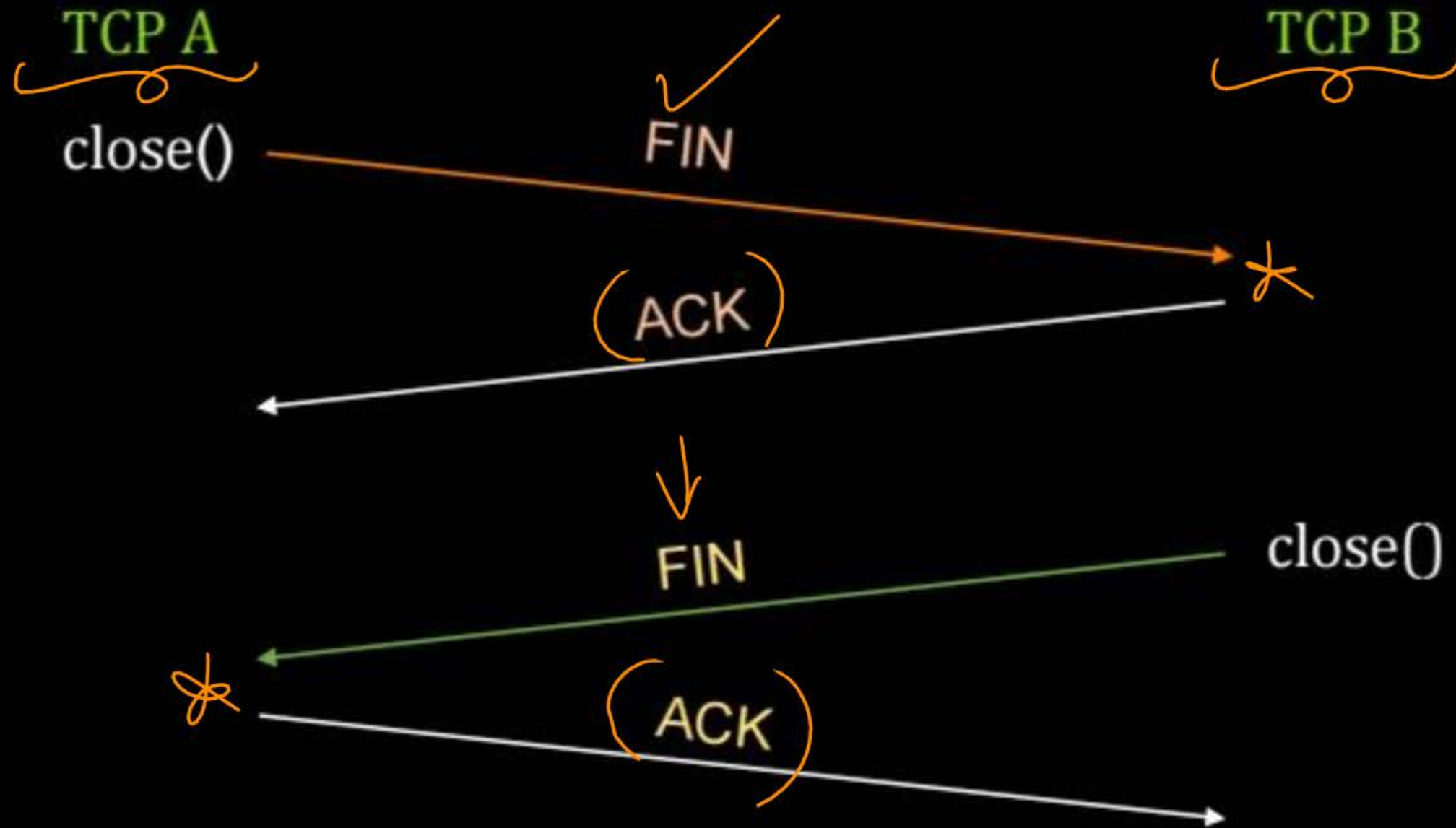
Topic : Close a Socket



```
ssize_t close ( int sockfd );
```



Topic : TCP Socket Programming





2 mins Summary



Topic

Socket API



THANK - YOU